The empirical recurrence for OEIS A233071, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A233071 records "Empirical recurrence of order 58" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 123$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 61$. Its last coefficient is $c_{58} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A233071 is "Number of nX6 0..3 arrays with no element x(i,j) adjacent to value 3-x(i,j) horizontally, antidiagonally or vertically, top left element zero, and 1 appearing before 2 in row major order.". It has offset 1 and begins

$$122, 9801, 952869, 95264272, 9622519979, 975239425635, 98950914937033, 10043490383783484, \ldots$$

The entry states:

> Empirical recurrence of order 58 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 07:51:09, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 123$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 123$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 246$ exact terms from $n = 3$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{58} c_i \, a(n-i),$$

c_1	161	c_{16}	−27528480427035	c_{31}	−26305727820951404	c_{46}	−119010284769
c_2	−7186	c_{17}	−76283209169108	c_{32}	−15063329846463046	c_{47}	82682349581
c_3	125305	c_{18}	171331918529552	c_{33}	23849084605746324	c_{48}	−28822004898
c_4	−913258	c_{19}	355001845678701	c_{34}	−3353844604448333	c_{49}	7323574465
c_5	1607106	c_{20}	−760160260451093	c_{35}	−8845579154747992	c_{50}	−1501554927
c_6	21260654	c_{21}	−1432430298353490	c_{36}	5694586513697128	c_{51}	249596223
c_7	−219200766	c_{22}	2924494658394752	c_{37}	−312261738946584	c_{52}	−34171005
c_8	901057167	c_{23}	4210071968533820	c_{38}	−1281814308403672	c_{53}	3821063
c_9	762960156	c_{24}	−9511977074496427	c_{39}	806041466047801	c_{54}	−340195
c_{10}	−25913367861	c_{25}	−7351665174337866	c_{40}	−248283511755412	c_{55}	24105
c_{11}	98262642671	c_{26}	22970851419103763	c_{41}	26559197467341	c_{56}	−1363
c_{12}	108712361520	c_{27}	3963310394029314	c_{42}	15980088981747	c_{57}	54
c_{13}	−1731416918650	c_{28}	−36943978129361938	c_{43}	−9939653821226	c_{58}	−1
c_{14}	1986940453414	c_{29}	11127679345627302	c_{44}	2542752519775		
c_{15}	14055927761613	c_{30}	35393064223413188	c_{45}	−232002466260		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 61$, and it is the recurrence OEIS A233071 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{58-1} - \cdots - c_{58}$. Its constant term is $-c_{58} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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